



Project Proposal For 5G Innovation Hackathon 2026

Project Title

5G-Enabled Smart Emergency Response System (SERS)

Leveraging Edge AI, IoT Surveillance & 5G Networks for Real-Time Public Safety Infrastructure

Faculty Mentor:

Sr. No	Name	Department	Institute
1	Dr. Sandeep Kumar Singh (Assistant Professor)	DoECE	NIT Hamirpur

Team Members:

Sr. No	Name	Roll No.	Department	Institute
1	Ashish Singh	23BEC025	DoECE	NIT Hamirpur
2	Harsh Aryan	23DEC014	DoECE	NIT Hamirpur
3	Harsh Verma	24BEE048	DoEE	NIT Hamirpur
4	Adarsh Kumar Singh	24BEC006	DoECE	NIT Hamirpur

1.Objective

The objective of the Smart Emergency Response System (SERS) is to build a real-time, autonomous public safety platform that utilizes 5G URLLC, Edge AI, and IoT-based surveillance to:

- *Detect accidents and critical incidents without human intervention*
- *Reduce emergency response time from >8 minutes to <2 minutes*
- *Enable sub-second alert generation (<300 ms end-to-end)*
- *Integrate ambulance dispatch, traffic control, and hospital systems*
- *Provide a scalable infrastructure for smart cities, highways, and industrial safety*

2.Summary of Proposal

The Smart Emergency Response System (SERS) is a next-generation, AI-driven public safety platform that eliminates the critical delay inherent in manual emergency reporting. By fusing 5G Ultra-Reliable Low Latency Communication (URLLC), distributed Edge AI, and IoT-enabled surveillance, SERS creates an autonomous, always-on emergency detection and response ecosystem.

Problem Statement

Current emergency response workflows rely on human-initiated calls, resulting in average response times exceeding 8 minutes. Every minute of delay in trauma care reduces survival rates by up to 10%. SERS addresses this gap with machine-driven detection and dispatch, targeting a response time of under 2 minutes.

SERS addresses the following critical gaps in current infrastructure:

- Eliminates dependence on bystander or victim-initiated emergency calls
- Delivers autonomous incident detection within milliseconds of an event
- Integrates ambulance dispatch, traffic signal control, and hospital pre-alerting into a unified automated pipeline
- Scales from individual intersections to city-wide smart infrastructure deployments

Performance Metric	Target Value
End-to-end alert latency	< 10 ms
Incident detection time (post-event)	< 80 ms per frame
Emergency response time	< 2 minutes (down from > 8 min)
Network reliability (URLLC)	99.999%
Estimated reduction in fatalities	~40%
Deployment scalability	10,000+ nodes

3. Detailed Project Proposal

SERS is designed as a distributed edge-cloud architecture with deep 5G integration across four functional layers. Each layer is independently scalable and communicates through standardized protocols.

3.1 Layer 1 — Intelligent Surveillance

The Intelligent Surveillance layer serves as the foundational data acquisition component of the system. It consists of high-resolution ONVIF-compliant IP cameras that continuously capture real-time video streams at 25–30 frames per second, with resolutions ranging from 1080p to 4K. These cameras are deployed across critical locations such as intersections, highways, and industrial zones. Optional sensors such as LiDAR and thermal cameras enhance the system's ability to operate in low-light conditions and provide depth and thermal information for improved incident detection. This layer is designed to integrate seamlessly with existing CCTV infrastructure, ensuring scalability and cost-effective deployment while acting as the primary source of visual data for downstream processing.

Component	Specification
Camera Standard	ONVIF-compliant IP cameras with RTSP streaming
Frame Rate	25–30 FPS
Resolution	1080p to 4K
Depth Sensing (Optional)	LiDAR modules for 3D incident analysis
Thermal Imaging (Optional)	Thermal cameras for nighttime and fire detection

- This layer captures real-time visual data using IP cameras deployed at roads, intersections, or industrial sites.
- It continuously streams video (RTSP) for monitoring without human involvement.
- Optional sensors like LiDAR and thermal cameras enhance detection in low-light or complex environments.
- It acts as the **data source** for the entire system.
- Designed to work with existing CCTV infrastructure for easy scalability.

3.2 Layer 2 — Edge AI Processing

The Edge AI Processing layer is responsible for performing real-time analysis of video data at the source, minimizing latency and reducing dependence on centralized cloud systems. This layer utilizes edge computing devices such as Raspberry Pi or NVIDIA Jetson Nano, accelerated by Google Coral TPU for efficient AI inference. The video frames are processed through a structured pipeline involving preprocessing, object detection using YOLOv8, and post-processing techniques like Non-Max Suppression. The system detects critical events such as vehicle collisions, pedestrian accidents, and fire incidents, and classifies them based on severity. By processing data locally with an inference latency of under 80 milliseconds, this layer ensures rapid decision-making and transmits only relevant event data instead of raw video, optimizing bandwidth usage.

Hardware

- Raspberry Pi 5 or NVIDIA Jetson Nano (primary compute)
- Google Coral TPU — 4 TOPS hardware acceleration for inferenceInference Pipeline
-

Step	Operation
1	Frame acquisition via OpenCV / GStreamer pipeline
2	Preprocessing: resize to 640x640, normalize pixel values
3	YOLOv8 inference (TFLite-optimized for edge hardware)
4	Post-processing: Non-Max Suppression, bounding box extraction
5	Event classification and severity scoring

Target inference latency: < 80 ms per frame

- This layer performs real-time AI inference directly on edge devices like Raspberry Pi or Jetson Nano.
- It processes video frames using models like YOLOv8 to detect accidents or anomalies.
- Preprocessing, detection, and classification happen locally to reduce latency.
- Only important events (not raw video) are sent forward, saving bandwidth.
- Ensures **fast decision-making (<80 ms)** without depending on the cloud.

3.3 Layer 3 — 5G Communication

The 5G Communication layer acts as the high-speed, low-latency bridge between the edge processing units and the cloud infrastructure. It utilizes a 5G Standalone (SA) network configured with Ultra-Reliable Low Latency Communication (URLLC) to ensure end-to-end latency below 10 milliseconds and reliability of up to 99.999%. This layer enables instantaneous transmission of structured alert data, including event type, location, and severity. It supports multiple communication protocols such as MQTT over TLS for lightweight messaging, REST APIs for system integration, and WebRTC for real-time video streaming. Network slicing ensures that emergency data is prioritized over regular traffic, making this layer critical for maintaining real-time responsiveness in mission-critical scenarios.

5G Parameter	Specification
Network Type	5G Standalone (SA)
Slice Type	URLLC
End-to-End Latency	< 10 ms
Throughput per Camera	15–25 Mbps
Reliability	99.999% (mission-critical)
Edge Integration	MEC (Multi-Access Edge Computing) nodes

- This layer transmits detected events using a dedicated 5G network optimized for low latency (URLLC).
- It ensures reliable and near-instant communication between edge devices and the cloud.
- Uses protocols like MQTT for alerts, REST APIs for control, and WebRTC for live video.
- Network slicing ensures emergency data is prioritized over regular traffic.
- Acts as the **high-speed bridge** connecting detection and response systems.

Protocols

- MQTT over TLS — lightweight IoT event messaging
- HTTP/REST APIs — control signal exchange with dispatch systems
- WebRTC — low-latency live video streaming to command centers

3.4 Layer 4 — Cloud Orchestration

The Cloud Orchestration layer functions as the central intelligence hub of the system, managing data processing, decision-making, and coordination of emergency responses. Built on a cloud-native architecture using services such as AWS IoT Core, AWS Lambda, DynamoDB, and Amazon S3, this layer processes incoming alerts through event-driven workflows. It employs advanced dispatch logic, including geo-fencing algorithms to identify the nearest available emergency units and priority-based queuing to handle multiple incidents simultaneously. Additionally, it stores historical data for analysis and system optimization. This layer ensures seamless integration with external systems such as ambulance services, traffic management systems, and hospitals, enabling efficient and automated response coordination.

AWS Service Stack

Service	Role
AWS IoT Core	Secure device communication and message routing
AWS Lambda	Serverless event processing and dispatch logic
Amazon DynamoDB	Real-time incident database with low-latency reads
Amazon S3	Long-term video evidence archival
API Gateway	External integrations with emergency services

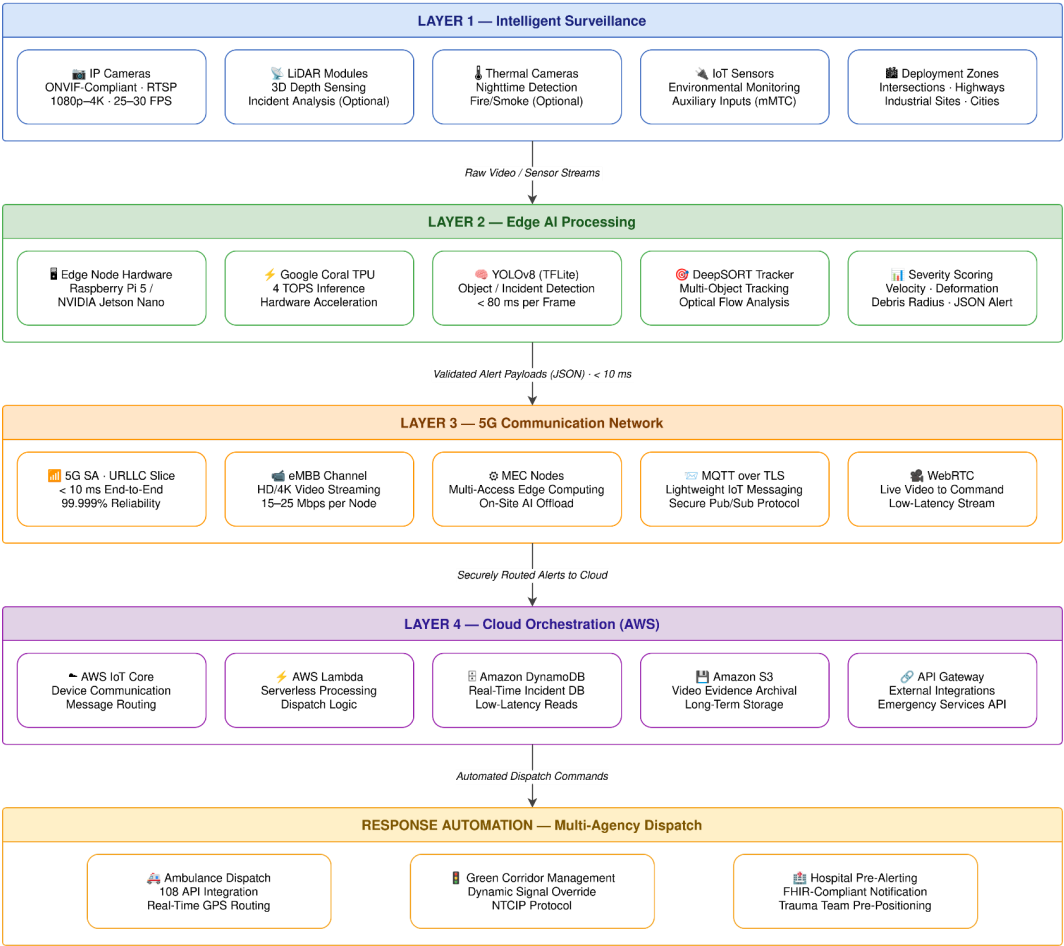
- This layer receives alerts and manages the overall response workflow.
- It processes events using serverless services (like AWS Lambda) and stores data in databases.
- It decides which ambulance or service to dispatch based on location and priority.
- Also handles long-term storage, analytics, and system monitoring.
- Acts as the **central brain** coordinating all emergency actions.

Dispatch Intelligence

- Geo-fencing algorithm to identify and notify the nearest available ambulance unit
- Priority-based queueing system for concurrent multi-incident scenarios
- Real-time resource allocation and conflict resolution

5G-Enabled Smart Emergency Response System (SERS)

End-to-End Architecture: Surveillance → Edge AI → 5G → Cloud → Response



Key Performance Targets

- Alert Latency: < 10 ms
- Detection: < 80 ms/frame
- Response Time: < 2 min
- Reliability: 99.999%
- Fatality Reduction: ~40%

Legend

- Surveillance
- Edge AI
- 5G Network
- Cloud (AWS)
- Response Automation

4. AI Detection & Analytics Engine

4.1 Detection Capabilities

The SERS AI engine is trained to detect a comprehensive set of critical incidents in real time:

- Vehicle collision detection (front, rear, and side impact)
- Vehicle rollover detection
- Pedestrian impact events
- Fire and smoke detection (visual and thermal fusion)
- Abnormal vehicular behaviour — sudden deceleration, wrong-way driving

4.2 Advanced Analytics

- Multi-object tracking using DeepSORT for persistent identity across frames
- Trajectory analysis via optical flow algorithms
- Behaviour anomaly detection using LSTM networks (planned for v2.0)

4.3 Severity Estimation Algorithm

Upon detecting an incident, SERS computes a severity score using a composite algorithm:

- Impact velocity (estimated from frame-to-frame object displacement)
- Object deformation proxy (bounding box aspect ratio change)
- Debris dispersion radius

Sample Alert Payload (JSON)

```
{
  "event_type": "collision",
  "severity": "critical",
  "confidence": 0.94,
  "location": "12.9716, 77.5946",
  "timestamp": "2025-04-20T14:32:01.204Z",
  "camera_id": "CAM-BLR-042",
  "dispatch_triggered": true
}
```

5. 5G Technology Utilization

The Smart Emergency Response System (SERS) leverages the full capabilities of 5G technology to enable real-time, reliable, and large-scale emergency management. It utilizes **Ultra-Reliable Low Latency Communication (URLLC)** to ensure that critical alerts, such as accident detection notifications, are transmitted within **sub-10 millisecond latency** with extremely high reliability (up to 99.999%), making it suitable for life-critical applications. The system also employs **enhanced Mobile Broadband (eMBB)** to support high-quality video streaming from surveillance cameras, enabling live monitoring and verification of incidents at data rates of 15–25 Mbps per node. Additionally, **massive Machine Type Communication (mMTC)** allows the system to connect and manage thousands of IoT devices, including cameras, LiDAR, and thermal sensors, simultaneously without network congestion. Advanced features such as **network slicing** provide a dedicated communication channel for emergency data, ensuring priority handling even during peak network usage. Furthermore, the integration of **Multi-access Edge Computing (MEC)** enables processing closer to the data source, reducing dependency on centralized cloud systems and further minimizing response time. Together, these 5G capabilities form the backbone of SERS, enabling fast, scalable, and highly reliable emergency response operations.

5G Capability	Use in SERS	Benefit
URLLC	Emergency alert transmission	Sub-10 ms, 99.999% reliable
eMBB	HD/4K video streaming	15–25 Mbps per node
mMTC	IoT sensor network (LiDAR, thermal)	Thousands of concurrent devices
Network Slicing	Dedicated emergency channel	Isolated from congestion
MEC / Edge	On-site AI inference	Cloud-independent real-time processing

- SERS uses **URLLC (Ultra-Reliable Low Latency Communication)** to ensure alerts are delivered within milliseconds.
- **eMBB** supports high-quality video streaming from cameras to control centers.
- **mMTC** enables connection of thousands of IoT devices like sensors and cameras simultaneously.
- **Network slicing** creates a dedicated emergency channel, avoiding congestion from normal users.
- **MEC (Multi-access Edge Computing)** allows AI processing closer to the source, reducing dependency on cloud.

6. Multi-Domain Use Cases

SERS is designed as a general-purpose AI safety platform. The following use cases are addressable within the same hardware and software stack:

Domain	Application
Emergency Response (Core)	Automated accident detection, ambulance dispatch, green-corridor traffic management
Smart Surveillance	Loitering/violence detection, crowd density analysis, perimeter security
Fire & Disaster	Early fire/smoke detection, flood and landslide monitoring
Traffic Management	Congestion detection, adaptive signal control, vehicle flow optimization
Industrial Safety	Hazard detection, worker compliance monitoring, chemical leak alerts
Drone Surveillance	Aerial incident verification, disaster area assessment before ground dispatch

- In **emergency response**, it detects accidents and automatically dispatches ambulances with optimized routing.
- In **smart surveillance**, it can monitor crowd behavior, detect violence, or ensure perimeter security.
- For **fire and disaster management**, it identifies smoke, fire, or environmental hazards early.
- In **traffic management**, it analyzes congestion and dynamically controls traffic signals.
- In **industrial safety**, it detects hazards, worker violations, or chemical leaks in real time.

7. Response Automation & Integration

Upon incident confirmation, SERS triggers a fully automated multi-agency response without human intervention:

- Ambulance Dispatch — integration with 108 emergency services API for direct unit assignment and GPS routing
- Green Corridor Management — dynamic traffic signal overrides via NTCIP protocol, clearing a path from the incident site to the nearest hospital
- Hospital Pre-Alerting — FHIR-compliant pre-notification to receiving hospitals, enabling trauma team pre-positioning before patient arrival

8. Societal & Commercial Impact

8.1 Societal Impact

- Estimated 40% reduction in road accident fatalities through faster response
- Faster trauma care initiation — every minute saved translates to measurably higher survival rates

- Enhanced urban safety for pedestrians, cyclists, and vulnerable road users
- Scalable public good infrastructure compatible with India's Smart Cities Mission

8.2 Commercial & Industrial Relevance

- Direct alignment with the global AI surveillance market (projected \$100B+ by 2030)
- Telecom operators can offer SERS as a managed 5G enterprise service on URLLC slices
- Industrial safety vertical — significant demand in manufacturing, oil & gas, and logistics
- Reusable platform architecture enables rapid deployment across new verticals

9. Scalability & Market Readiness

Dimension	Detail
Technology Readiness	TRL 6–7 (Prototype complete, Pilot-ready)
Node Scalability	Supports 10,000+ deployment nodes on cloud-native stack
Legacy Compatibility	Compatible with existing CCTV infrastructure (ONVIF/RTSP)
Infrastructure Model	Cloud-native microservices; horizontal auto-scaling
Deployment Modes	City-wide, highway corridor, industrial campus

10. Conclusion

The Smart Emergency Response System (SERS) presents a modern, technology-driven solution to the critical challenge of delayed emergency response. By integrating **Edge AI, 5G communication, IoT surveillance, and cloud-based orchestration**, the system enables real-time accident detection and automated response within seconds.

Unlike traditional methods, SERS eliminates reliance on manual reporting by using intelligent edge processing and ultra-low latency 5G networks to ensure rapid communication and decision-making. The system not only improves emergency response efficiency but also supports multiple applications such as smart surveillance, traffic management, and disaster monitoring.

With its scalable and modular architecture, SERS can be deployed across cities, highways, and industrial environments. By reducing response time from minutes to seconds, it has the potential to significantly lower fatalities and enhance public safety, making it a future-ready solution for smart infrastructure.

SERS is not a prototype concept — it is a deployable, standards-compliant platform with a clear path from pilot to city-scale production.